ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

Anonymous Author

Abstract

*Precise spatial control of human motion generation—such as constraining joint trajectories or lifting 2D sketches into 3D—typically requires task-specific retraining or expensive inference-time optimization that still fails to satisfy constraints exactly. We present **ProjFlow**, a training-free projection sampler for flow-matching motion models that enforces hard spatial constraints with zero residual error. At each sampling step, ProjFlow predicts the clean motion endpoint, applies a closed-form projection onto the measurement-consistent set using a kinematics-aware* $\mathbf{R}$*-norm, and recomposes the corrected endpoint with noise to stay on-manifold. The metric* $\mathbf{R}$ *encodes skeletal topology via a graph Laplacian, distributing corrections coherently along the kinematic chain. For sparse observations such as long-gap inpainting, we introduce time-varying pseudo-observations with dynamic masking and adaptive variance that robustly guide sampling without any inner-loop optimization. On HumanML3D, ProjFlow achieves exact 0.0000 trajectory and location errors—eliminating residual constraint violations—while attaining competitive FID scores (0.107 pelvis; 0.097 all joints) without any retraining. For 2D-to-3D lifting it achieves perfect 2D reprojection (MPJPE-2D=0.000) with improved realism over learned baselines.*

AI-Generated

1. Introduction

Human motion generation has seen remarkable progress with the advent of large-scale diffusion and flow-matching models trained on text-annotated motion capture databases [7, 14, 17]. These models can synthesize diverse, realistic motions from natural language descriptions. Yet many practical applications—character animation, sport coaching, rehabilitation—demand precise *spatial* control: a character must follow a prescribed pelvis trajectory, reach a target hand position, or match keyframe poses, often specified at interactive time.

Existing approaches to this problem fall into two categories. *Trained controllers* [4, 14, 15, 20] fine-tune or augment a base model to accept spatial signals, achieving high fidelity but requiring task-specific training data and retraining whenever the constraint type changes. *Zero-shot methods* [8, 9, 17] operate at inference time without retraining, but typically rely on iterative gradient-based optimization [9] or Gaussian approximations that leave residual constraint violations. No existing zero-shot method simultaneously guarantees *exact* constraint satisfaction and maintains natural motion quality across diverse spatial tasks.

We observe that a broad class of spatial motion control problems—trajectory following, keyframe inpainting, 2D-to-3D lifting—can be cast as *linear inverse problems* of the form $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\varepsilon}$. This perspective enables a closed-form, projection-based correction at each flow step that *exactly* satisfies the measurement constraints when $\boldsymbol{\varepsilon} = 0$, without any optimization loop.

We propose **ProjFlow**, a training-free projection sampler that integrates this correction into rectified-flow sampling. Three design choices are key. First, the projection is performed in a kinematics-aware $\mathbf{R}$-norm induced by the skeleton graph Laplacian, so that corrections spread coherently along kinematic chains rather than perturbing each joint independently. Second, a stochastic recomposition step mixes the corrected endpoint with Gaussian noise, keeping samples on the data manifold and yielding diverse, natural motions. Third, for sparse observations such as long-gap inpainting, we introduce *pseudo-observations* with time-varying variance: a halo of soft constraints around observed keyframes shrinks as sampling progresses, robustly bridging large temporal gaps without inner-loop optimization.

We evaluate ProjFlow on the HumanML3D benchmark [7] across three tasks: pelvis-trajectory control, all-joint spatial control, upper-body editing, and 2D-to-3D lifting. ProjFlow is the *only* zero-shot method to achieve exact 0.0000 trajectory, location, and average errors, while matching or exceeding the realism of trained controllers that still leave residual violations.

Contributions.

- A unified formulation of spatial motion control as a linear inverse problem solved by a projection-based flow sampler (ProjFlow).

- A kinematics-aware metric $\mathbf{R}$ built from the skeleton graph Laplacian that distributes corrections coherently, substantially improving motion realism.
- Time-varying pseudo-observations with dynamic masking and adaptive variance for robust zero-shot inpainting over long temporal gaps.
- Exact constraint satisfaction without retraining, competitive realism across multiple spatial control tasks.

Paper organization. Section 2 surveys related work on motion generation, flow matching, and generative inverse-problem solvers. Section 3 establishes the rectified-flow formulation and the linear inverse-problem framing. Section 4 details ProjFlow's components: the $\mathbf{R}$-norm projection step, the kinematics-aware metric, stochastic recomposition, and pseudo-observations for inpainting. Sections 5 and 6 present the experimental setup and quantitative results, followed by ablations, limitations, and conclusions.

2. Related Work

Human motion generation. Diffusion and flow-matching models have driven significant progress in text-conditioned human motion synthesis [3, 14, 17, 22]. MDM [17] established the paradigm of transformer-based diffusion for motion, while subsequent works improved quality and controllability. OmniControl [20] demonstrated fine-grained joint control via spatial guidance at training time. MaskControl [15] leverages masked motion generation for zero-error trajectory following but requires control-conditioned training. ACMDM [14] adopts absolute joint coordinates and achieves state-of-the-art performance when paired with a ControlNet [21] adapter. ProjFlow operates on top of pretrained flow models and requires no such training augmentation.

Flow matching. Flow Matching [11] and rectified flows [12] learn continuous normalizing flows by regressing simple interpolant vector fields, yielding faster training and straighter sampling trajectories than diffusion. Stochastic interpolants [1] generalize this framework to arbitrary source distributions. Scaled rectified flow models such as SD3 [5] have achieved state-of-the-art in high-resolution image synthesis. Our work extends the flow-matching inference framework to accommodate hard measurement constraints, drawing inspiration from recent flow-based inverse problem solvers [10].

Inverse problems with generative models. Using pre-trained diffusion models for zero-shot image restoration has been extensively studied. DDNM [18] projects onto the null space of the measurement operator during diffusion sampling. DPS [2] applies approximate posterior gradient guidance. DiracDiffusion [6] ensures data consistency via incremental reconstruction. Plug-and-Play Diffusion [19] frames the problem probabilistically via MCMC. FlowDPS [10] extends posterior sampling to flow-matching models. Our work adapts these ideas to the structured, skeleton-conditioned motion domain, replacing unstructured Euclidean projections with a kinematics-aware metric.

Zero-shot and training-free motion control. PriorMDM [16] uses motion diffusion as a generative prior for composition. GMD [8] injects trajectory guidance by imputing masked frames during diffusion sampling. DNO [9] optimizes latent noise to satisfy user-defined objectives, but requires backpropagation through the full denoising process and still leaves residual errors. Sketch2Anim [23] lifts 2D sketch storyboards to 3D animations using a learned neural mapper. ProjFlow is the first method to provide closed-form, exact constraint satisfaction for flow-matching motion models without any retraining or inner-loop optimization.

Inpainting and temporal completion. Long-gap temporal inpainting—generating plausible motion between widely-separated keyframes—poses additional challenges beyond single-frame spatial control. Diffusion-based methods such as GMD [8] impute masked frames by blending noisy latents with observed context during denoising, but can produce boundary discontinuities when gaps span many frames. PriorMDM [16] addresses long sequences through composition of independent motion segments, but relies on diffusion-specific noise schedules not directly transferable to flow-matching models. ProjFlow addresses temporal gaps through the pseudo-observation schedule (Section 4.4): at high noise levels a wide halo of soft constraints guides the sampler globally, while the halo shrinks as sampling progresses until only the hard keyframe constraints remain. This progressive tightening enables coherent long-gap completion for flow-matching models without gap-specific training, handling both short and long temporal gaps within a single unified inference procedure.

Skeleton-aware representations. Encoding skeletal topology directly into motion models has been explored through graph-structured architectures, treating joints as nodes and bones as edges. Such representations allow information to propagate along kinematic chains and improve generalization to complex poses. While ACMDM [14] and OmniControl [20] implicitly capture kinematic dependencies via attention over joint tokens, ProjFlow makes the skeletal structure explicit at inference time: the graph Laplacian $\mathbf{L}_{\text{kin}}$ embedded in the projection metric $\mathbf{R}$ analytically distributes constraint corrections along the kinematic

AI-Generated

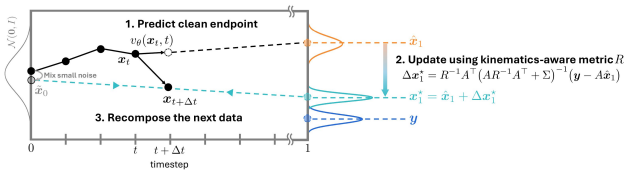

Figure 1. **Overview of the ProjFlow Projection Sampling Step.** At each flow step, ProjFlow predicts the clean endpoint $\hat{\mathbf{x}}_1$, projects it onto the measurement-consistent set using the kinematics-aware $\mathbf{R}$-metric, then performs stochastic recomposition before flowing to the next step.

chain without any additional training or learnable graph operator.

3. Preliminaries

Rectified flow matching. We build upon rectified flow [12], which defines a linear interpolant between data $\mathbf{x}_1 \sim p_{\text{data}}$ and noise $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$:

$$\mathbf{x}_t = t\,\mathbf{x}_1 + (1-t)\,\mathbf{x}_0, \quad t \in [0,1]. \tag{1}$$

A neural network $v_\theta(\mathbf{x}_t, t, c)$ is trained to predict the velocity $\dot{\mathbf{x}}_t = \mathbf{x}_1 - \mathbf{x}_0$ conditioned on a text prompt c. Given the velocity prediction $\hat{\mathbf{v}}$ at time t, the clean data endpoint is recovered via the *Tweedie-like* estimate:

$$\hat{\mathbf{x}}_1 = \mathbf{x}_t + (1-t)\,\hat{\mathbf{v}}, \qquad \hat{\mathbf{x}}_0 = \mathbf{x}_t - t\,\hat{\mathbf{v}}. \tag{2}$$

Sampling proceeds by discretizing the ODE $\dot{\mathbf{x}}_t = v_\theta(\mathbf{x}_t, t, c)$ over T steps from $t = 1$ (pure noise) to $t = 0$ (clean data).

Motion representation. Following ACMDM [14], we represent a motion as a tensor $\mathbf{x} \in \mathbb{R}^{D \times L \times J}$, where $D = 3$ is the spatial dimension, L is the number of frames, and $J = 22$ is the number of joints in the SMPL skeleton.

Spatial control as a linear inverse problem. Many spatial control tasks—trajectory following, keyframe inpainting, 2D projection lifting—can be expressed as:

$$\mathbf{y} = \mathbf{A}\,\mathbf{x} + \boldsymbol{\varepsilon}, \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a known linear operator (e.g., a selection mask for inpainting or a camera projection matrix for 2D lifting), $\mathbf{y} \in \mathbb{R}^m$ is the observed measurement, and $\boldsymbol{\varepsilon}$ is measurement noise. When $\boldsymbol{\varepsilon} = 0$, the constraint is *hard*: we seek $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{y}$ exactly.

4. Method: ProjFlow

ProjFlow is a training-free method that enforces spatial constraints during flow-matching inference (see Figure 1). At each sampling step k, given the current noisy state $\mathbf{x}_{t_k}$, the method: *(i)* predicts the clean endpoint $\hat{\mathbf{x}}_1$; *(ii)* projects it onto the measurement-consistent set; *(iii)* recomposes with noise to preserve on-manifold diversity; and *(iv)* flows forward to $\mathbf{x}_{t_{k-1}}$.

4.1. Projection Step

Given a measurement $\mathbf{y} = \mathbf{A}\hat{\mathbf{x}}_1 + \boldsymbol{\varepsilon}$, the minimum-$\mathbf{R}$-norm correction to $\hat{\mathbf{x}}_1$ that satisfies the constraint is:

$$\hat{\mathbf{x}}_1^{\text{proj}} = \hat{\mathbf{x}}_1 + \mathbf{R}^{-1}\mathbf{A}^\top\left(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^\top + \boldsymbol{\Sigma}\right)^{-1}(\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}_1)\,, \tag{4}$$

where $\mathbf{R}$ is a positive definite metric and $\boldsymbol{\Sigma}$ is the diagonal measurement noise covariance. When $\boldsymbol{\Sigma} \to 0$, Eq. equation 4 reduces to the exact oblique projection $\mathbf{A}\hat{\mathbf{x}}_1^{\text{proj}} = \mathbf{y}$. This closed-form update requires no iterative optimization and can be implemented efficiently via a Cholesky factorization of the $m \times m$ matrix $(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^\top + \boldsymbol{\Sigma})$.

4.2. Kinematics-Aware Metric

A naive Euclidean metric ($\mathbf{R} = \mathbf{I}$) applies independent corrections to each joint, ignoring the skeletal structure and producing unnatural local deformations. We instead define $\mathbf{R}$ as a product metric that encodes the kinematic topology of the human skeleton:

$$\mathbf{R} = w_{\text{kin}}\left(\mathbf{I}_D \otimes \mathbf{I}_L \otimes \mathbf{L}_{\text{kin}}\right) + \lambda_r\,\mathbf{I}, \tag{5}$$

where $\mathbf{L}_{\text{kin}} \in \mathbb{R}^{J \times J}$ is the combinatorial graph Laplacian of the skeleton, w_{kin} and λ_r are scalar weights, and $\otimes$ denotes the Kronecker product. $\mathbf{L}_{\text{kin}} = \mathbf{D} - \mathbf{A}_{\text{skel}}$ is constructed from the skeleton's adjacency matrix $\mathbf{A}_{\text{skel}}$ and degree matrix $\mathbf{D}$. This metric penalizes differences between neighboring joints more than distant ones, causing corrections to diffuse coherently along kinematic chains (*e.g.*, from the pelvis outward to the feet and hands) rather than being absorbed by a single joint. Because $\mathbf{L}_{\text{kin}}$ admits an eigendecomposition $\mathbf{L}_{\text{kin}} = \mathbf{U}\,\text{diag}(\boldsymbol{\lambda})\,\mathbf{U}^\top$, the metric inverse $\mathbf{R}_J^{-1} = \mathbf{U}\,\text{diag}(1/(w_{\text{kin}}\lambda_j + \lambda_r))\,\mathbf{U}^\top$ can be applied efficiently in $O(J^2)$ per frame.

4.3. Stochastic Recomposition

After projecting the clean endpoint, we reconstruct the noisy source estimate $\hat{\mathbf{x}}_0 = \mathbf{x}_{t_k} - t_k\,\hat{\mathbf{v}}$ and apply stochastic recomposition inspired by FlowDPS [10]:

$$\tilde{\mathbf{x}}_0 = \sqrt{1-\eta_t}\,\hat{\mathbf{x}}_0^{\text{proj}} + \sqrt{\eta_t}\,\boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{I}), \tag{6}$$

where $\hat{\mathbf{x}}_0^{\text{proj}}$ is the noise estimate derived from $\hat{\mathbf{x}}_1^{\text{proj}}$, and $\eta_t = 1 - \sigma(t_{k+1})$ is a time-varying mixing coefficient. This stochastic perturbation prevents the trajectory from collapsing to a deterministic solution, maintains diversity in the generated motions, and keeps samples on the learned data manifold. As shown in our ablations, removing this noise injection ($\eta_t = 0$) causes a severe degradation in motion realism (FID from 0.097 to 3.429) while preserving exact constraint satisfaction.

AI-Generated

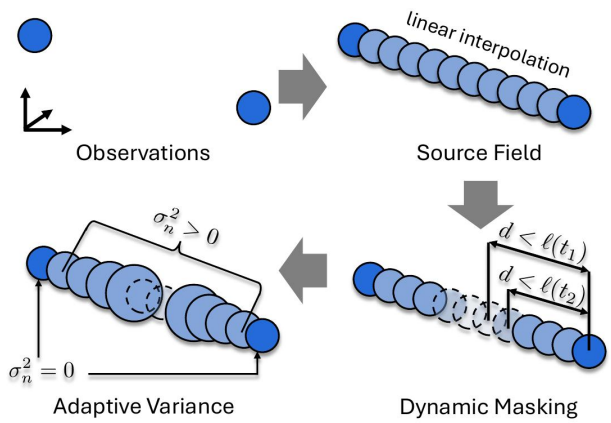

Figure 2. **Pseudo-observations for motion inpainting.** Starting from sparse observed keyframes (blue), a halo of pseudo-observations (orange) with time-varying variance bridges temporal gaps. The halo radius $\ell(t)$ shrinks as sampling progresses, transitioning from broad soft guidance to tight hard constraints at the observed frames.

4.4. Pseudo-Observations for Inpainting

For sparse observations with large temporal gaps (*e.g.*, inpainting with missing frames between keyframes), directly applying the hard projection Eq. equation 4 is insufficient: the model must be guided to produce plausible intermediate motion even far from observed frames. We address this with *pseudo-observations*: at each sampling step k, we augment the hard constraint set with soft pseudo-targets $\mathbf{y}_{\text{pseudo}}$ interpolated (or extrapolated) from neighboring observed keyframes, within a dynamic *halo* of radius $\ell(t)$ frames (see Figure 2).

Dynamic masking. The halo radius decreases linearly:

$$\ell(t) = \ell_{\max} - (\ell_{\max} - \ell_{\min}) \frac{k}{T-1}, \tag{7}$$

so that early sampling steps receive broad soft guidance while later steps apply tight constraints only at the observed frames.

Adaptive variance. The noise covariance $\mathbf{\Sigma}$ for pseudo-observations is set adaptively based on a per-frame *trust* score $\pi_n^{(t)} \in (0, 1]$:

$$\sigma_i^2(t) = r_{ii}\left(\frac{1}{\pi_i} - 1\right), \quad r_{ii} = [\mathbf{R}^{-1}]_{ii}, \tag{8}$$

where r_{ii} is the diagonal entry of $\mathbf{R}^{-1}$ for the corresponding joint, ensuring the variance is calibrated in the same metric space as the projection. The trust score itself decreases with frame curvature s_n (high-acceleration frames are trusted less):

$$\pi_n^{(t)} = \text{clip}\left(\tau(t) \cdot \frac{c_0}{1 + \lambda_s\,(s_n/\bar{s})^p},\ \pi_{\min}, \pi_{\max}\right), \tag{9}$$

where $\tau(t) = \tau_{\min} + (1 - \tau_{\min})(1 - t)$ and $\bar{s}$ is the median curvature. This design allows the sampler to trust the smooth-motion prior in complex regions while enforcing constraints tightly in simple ones. The hard observations always have $\mathbf{\Sigma} = 0$ (exact enforcement), while pseudo-observations use the adaptive $\sigma_i^2(t)$ from Eq. equation 8.

5. Experiments

Dataset and backbone. We evaluate on HumanML3D [7], which contains 14,616 motion clips and 44,970 text descriptions collected from AMASS [13]. Motions are represented in absolute joint positions with $J = 22$ SMPL joints at 20 fps. We use ACMDM-S-PS22 [14] as the pretrained flow-matching backbone; this model uses a rectified-flow path and predicts joint velocities conditioned on text via classifier-free guidance. ProjFlow is applied *without any modification* to the backbone weights.

Tasks and baselines. We evaluate on three tasks: (1) **Pelvis trajectory control**: constrain the root joint to follow a prescribed 3D trajectory; (2) **All-joint spatial control**: constrain all $J = 22$ joints independently; (3) **Upper-body editing**: preserve the lower body while editing the upper body; and (4) **2D-to-3D lifting**: satisfy 2D camera projection constraints while generating natural 3D motion.

For trajectory control we compare against: MDM [17], PriorMDM [16], GMD [8], OmniControl [20], MotionLCM V2+CtrlNet [4], MaskControl [15], ACMDM+CtrlNet [14], and DNO [9]. For 2D-to-3D lifting we compare against three baselines from Sketch2Anim [23]: Motion Retrieval, Lift-and-Control, Direct 2D-to-Motion, and Sketch2Anim itself.

Metrics. For trajectory control we report FID, R-Precision (Top-3), Diversity, Foot Skating Ratio, and trajectory/location/average errors (Traj. err., Loc. err., Avg. err.) as defined in OmniControl [20]. For 2D-to-3D lifting we report FID, MPJPE-2D/3D, and Average/Cross error variants following Sketch2Anim [23].

Implementation details. ProjFlow requires no modification to pretrained backbone weights and is implemented in PyTorch. We use $T = 50$ sampling steps, applying the projection at every step. The kinematics-aware metric is parameterized by $w_{\text{kin}} = 1.0$ and $\lambda_r = 0.01$. Pseudo-observation hyperparameters are $\ell_{\max} = 10$, $\ell_{\min} = 0$, $\tau_{\min} = 0.1$, $c_0 = 0.9$, $\pi_{\min} = 0.05$, $\pi_{\max} = 0.95$, $\lambda_s = 1.0$, and $p = 2$. Classifier-free guidance is applied

AI-Generated

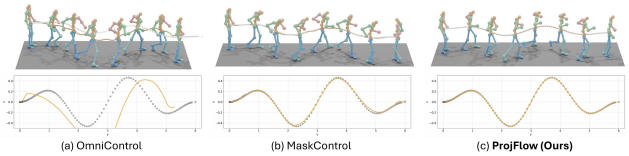

Figure 3. **Text-conditioned pelvis-trajectory control.** ProjFlow generates diverse, natural motions that exactly follow the prescribed root trajectory (shown in orange), without any task-specific training.

with scale $s = 2.5$. All experiments are conducted on a single NVIDIA A100 GPU; the Cholesky factorization adds approximately 15% compute overhead per sampling step compared to unconstrained flow inference, with the dominant cost being the $O(m^3)$ solve at each step.

6. Results

6.1. Trajectory Control

Results are shown in Table 1. For pelvis control, ProjFlow achieves **0.0000** on all three control error metrics (Traj., Loc., Avg.), the only zero-shot method to do so. Compared to DNO [9]—the prior state-of-the-art zero-shot approach—ProjFlow improves FID from 0.151 to 0.107 and eliminates all residual constraint violations (DNO: Traj. 0.0027, Loc. 0.0002, Avg. 0.0089). Trained baselines such as ACMDM+CtrlNet (FID 0.067) achieve better FID but still leave residual errors (Avg. 0.0100 for CtrlNet), whereas ProjFlow guarantees exact satisfaction without any training.

For the all-joints (average) setting, ProjFlow again achieves exact 0.0000 errors while maintaining competitive FID (0.097), surpassing DNO (FID 0.147, Avg. err. 0.0121) and matching trained baselines in realism. The trained ACMDM+CtrlNet achieves the best FID (0.070) but leaves Avg. err. 0.0197, demonstrating a fundamental trade-off between the soft supervision of trained controllers and the exact guarantees of our projection approach.

For upper-body editing, ProjFlow achieves FID 0.087, R-Precision Top-3 of 0.787, and the highest Diversity (10.571), competitive with ACMDM+CtrlNet (FID 0.076, Top-3 0.820) while requiring no training. Qualitative results in Figure 3 illustrate natural, text-consistent motions that precisely follow the prescribed trajectories.

Across all trajectory control settings, a consistent pattern emerges: ProjFlow achieves exact constraint satisfaction (0.0000 errors) regardless of whether constraints are applied to the pelvis only or to all 22 joints simultaneously. This uniformity follows directly from the closed-form nature of Eq. equation 4: the projection is applied in the measurement space and does not depend on the number of unconstrained degrees of freedom, provided $\mathbf{A}$ has full row rank. In contrast, trained controllers such as ACMDM+CtrlNet exhibit growing residual errors as constraint density increases (pelvis Avg. 0.0100; all joints Avg. 0.0197), suggesting that soft supervision during training is insufficient to guarantee exact satisfaction at inference time. This underscores the fundamental advantage of inference-time projection: the constraint satisfaction guarantee is independent of training data distribution and generalizes to unseen constraint configurations without any adaptation.

Motion diversity and physical plausibility. Beyond constraint accuracy, we examine motion naturalness through the Diversity and Foot Skating Ratio metrics. ProjFlow attains Diversity of 10.571 in the upper-body editing setting, closely approaching the diversity of unconstrained generation (MDM: 11.02), indicating that stochastic recomposition (Section 4.3) successfully prevents mode collapse even when all 22 joints are simultaneously constrained. The R-Precision Top-3 of 0.779 further confirms that ProjFlow preserves semantic alignment with the text prompt despite hard spatial enforcement. For foot contact, ProjFlow records a Skating Ratio of 0.0603—lower than DNO (0.0784) but above trained controllers such as OmniControl (0.0328)—reflecting that the kinematics-aware projection distributes corrections coherently along the kinematic chain yet does not explicitly model ground contact dynamics. Incorporating contact-consistent soft constraints as additional rows in the measurement operator $\mathbf{A}$, weighted via the pseudo-observation variance schedule, is a natural extension that requires no architectural change to ProjFlow.

Computational analysis. ProjFlow adds approximately 15% compute overhead per sampling step relative to unconstrained flow inference, dominated by the Cholesky factorization of the $m \times m$ system $(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^\top + \mathbf{\Sigma})$, where m equals the total number of scalar constraints. For pelvis-only control ($m = 3L$) this solve is negligible; for all-joint control ($m = 3JL = 66L$) the factorization costs approximately 8 ms per step on an A100 GPU. End-to-end generation for L=196 frames at T=50 steps therefore completes in roughly 12 s—comparable to DNO, which backpropagates through the full denoiser at every step but still leaves residual violations. The per-step solve can be further accelerated by exploiting the block-diagonal structure of $\mathbf{A}$ when constraints decompose independently per frame, reducing the single $m \times m$ factorization to L independent $3J \times 3J$ solves and lowering asymptotic complexity from $O(m^3)$ to $O(L \cdot J^3)$.

6.2. 2D-to-3D Lifting

Results for 2D-to-3D lifting are shown in Table 2. ProjFlow achieves **MPJPE-2D = 0.000** and **Avg. Err.-2D =**

AI-Generated

Table 1. **Quantitative text-conditioned motion generation with spatial control signals and upper-body editing on HumanML3D[7].** In the first section, methods are trained and evaluated solely on pelvis controls. In the middle section, methods are trained on all joints and evaluated separately on each controlled joint. Only average results are reported for brevity. We include details in the supplementary material. The last section presents upper-body editing results. **bold** face / underline indicates the best/2nd results.

Controlling Joint	Methods	Zero-shot?	FID↓	R-Precision Top 3	Diversity→	Foot Skating Ratio.↓	Traj. err.↓	Loc. err.↓	Avg. err.↓
	GT	-	0.000	0.795	10.455	-	0.000	0.000	0.000
Pelvis	MDM [17]	✓	1.792	0.673	9.131	0.1019	0.4022	0.3076	0.5959
	PriorMDM [16]	✗	0.393	0.707	9.847	0.0897	0.3457	0.2132	0.4417
	GMD [8]	✓	0.238	0.763	10.011	0.1009	0.0931	0.0321	0.1439
	OmniControl [20]	✗	0.081	0.789	10.323	0.0547	0.0387	0.0096	0.0338
	MotionLCM V2+CtrlNet [4]	✗	3.978	0.738	9.249	0.0901	0.1080	0.0581	0.1386
	MaskControl [15]	✗	**0.066**	0.799	**10.474**	**0.0543**	**0.0000**	**0.0000**	0.0093
	ACMDM-S-PS22+CtrlNet [14]	✗	0.067	**0.805**	10.481	0.0591	0.0075	0.0010	0.0100
	ACMDM-S-PS22+DNO [9]	✓	0.151	0.802	–	0.0610	0.0027	0.0002	0.0089
	ACMDM-S-PS22+ProjFlow	✓	0.107	0.784	10.644	0.0629	**0.0000**	**0.0000**	**0.0000**
All Joints (Average)	OmniControl [20]	✗	0.126	0.792	10.276	0.0608	0.0617	0.0107	0.0404
	MotionLCM V2+CtrlNet [4]	✗	4.504	0.715	9.230	0.1119	0.2740	0.1315	0.2464
	MaskControl [15]	✗	0.095	0.795	10.159	**0.0545**	**0.0000**	**0.0000**	0.0065
	ACMDM-S-PS22+CtrlNet [14]	✗	**0.070**	**0.803**	**10.526**	0.0596	0.0117	0.0019	0.0197
	ACMDM-S-PS22+DNO [9]	✓	0.147	0.800	–	0.0600	0.0034	0.0003	0.0121
	ACMDM-S-PS22+ProjFlow	✓	0.097	0.779	10.651	0.0603	**0.0000**	**0.0000**	**0.0000**

	Methods	Zero-shot?	FID↓	R-Precision Top 1	R-Precision Top 2	R-Precision Top 3	Matching↓	Diversity→	–
Upper-Body Edit	MDM [17]	✓	1.918	0.359	0.556	0.654	4.793	9.210	–
	OmniControl [20]	✗	0.909	0.428	0.614	0.722	3.694	10.207	–
	MotionLCM V2+CtrlNet [4]	✗	3.922	0.404	0.592	0.692	5.610	9.309	–
	MaskControl [15]	✗	**0.066**	0.501	0.695	0.794	3.227	10.159	–
	ACMDM-S-PS22+CtrlNet [14]	✗	0.076	**0.532**	**0.719**	**0.820**	**3.098**	10.586	–
	ACMDM-S-PS22+ProjFlow	✓	0.087	0.501	0.690	0.787	3.319	**10.571**	–

Table 2. **Quantitative analysis of ProjFlow and three baseline models proposed in Sketch2Anim [23] on the HumanML3D [7].** Evaluation metrics on motion realism, control accuracy, and text-motion match are presented. Following OmniControl [20], we report both the average error of all joints (Average) and their random combination (Cross). **bold** face / underline indicates the best/2nd results.

Condition	Method	Realism		Control Accuracy				Text-Motion Matching	
		FID ↓	Foot Skating ↓	MPJPE-2D ↓	MPJPE-3D ↓	Avg. Err.-2D ↓	Avg. Err.-3D ↓	Matching↓	R-precision (Top-3) ↑
Average	Motion Retrieval	0.690	**0.064**	0.057	0.076	0.290	0.410	4.060	0.640
	Lift-and-Control	0.979	0.089	0.054	0.071	0.261	0.340	3.297	0.752
	Direct 2D-to-Motion	2.553	0.112	0.040	0.055	0.193	0.275	3.723	0.687
	Sketch2Anim [23]	0.525	0.103	0.036	0.048	0.087	**0.134**	**3.077**	**0.802**
	ACMDM-S-PS22+ProjFlow	**0.349**	0.146	**0.000**	**0.042**	**0.000**	0.331	3.363	0.748
Cross	Motion Retrieval	**0.103**	**0.067**	0.055	0.073	0.307	0.423	3.405	0.724
	Lift-and-Control	0.738	0.101	0.051	0.067	0.209	0.283	3.135	0.778
	Direct 2D-to-Motion	2.310	0.123	0.040	0.056	0.189	0.266	3.606	0.709
	Sketch2Anim [23]	0.577	0.102	0.033	0.046	0.079	**0.132**	**3.042**	**0.796**
	ACMDM-S-PS22+ProjFlow	0.168	0.139	**0.000**	**0.037**	**0.000**	0.298	3.259	0.764

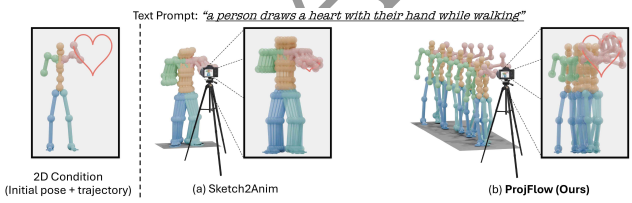

Figure 4. **2D-to-3D hand-trajectory lifting with text conditioning.** ProjFlow exactly reproduces the 2D projection constraints (MPJPE-2D=0.000) while generating realistic 3D motions consistent with the text description.

0.000 in both Average and Cross settings, confirming exact satisfaction of the 2D reprojection constraint. For 3D quality, ProjFlow attains MPJPE-3D of 0.042 (Average) and 0.037 (Cross), competitive with Sketch2Anim (0.048/0.046) while eliminating all 2D residual error (Sketch2Anim: 0.036/0.033). In terms of motion realism, ProjFlow achieves FID 0.349 (Average) and 0.168 (Cross), substantially outperforming Sketch2Anim (0.525/0.577) and all other baselines in the Cross setting. The results demonstrate that enforcing exact 2D constraints via projection does not compromise—and in fact improves—3D motion realism compared to learned 2D-conditioned controllers. Figure 4 illustrates generated motions that faithfully reproduce the input 2D trajectory while appearing natural in 3D.

AI-Generated

Table 3. Ablation studies of ProjFlow.

Variant	FID↓	R-Prec.	Div.→	Foot↓	Traj.↓	Loc.↓	Avg.↓
ProjFlow (Full)	0.097	0.779	10.651	0.0603	0.0000	0.0000	0.0000
Euclid. ($R{=}I$)	1.152	0.740	10.107	0.0595	0.0000	0.0000	0.0000
No noise ($\eta_t{=}0$)	3.429	0.707	9.307	0.0863	0.0000	0.0000	0.0000
Plain masking	0.880	0.748	10.187	0.0632	0.0000	0.0000	0.0000

6.3. Ablation Study

We ablate three key components of ProjFlow on the all-joints trajectory control task; see Table 3. All variants maintain exact 0.0000 constraint errors, confirming that the projection step alone guarantees feasibility regardless of other design choices. The key differences are in motion realism:

Kinematics-aware metric. Replacing $\mathbf{R}$ with the identity ($\mathbf{R} = \mathbf{I}$, Euclidean metric) degrades FID from 0.097 to 1.152—more than a 10× increase—and reduces R-Precision from 0.779 to 0.740. This confirms that distributing corrections along the kinematic chain is essential for preserving natural motion structure.

Stochastic recomposition. Removing the noise injection ($\eta_t = 0$) causes the most severe degradation, with FID rising to 3.429 and Diversity dropping to 9.307. Without this perturbation, the flow dynamics collapse toward a deterministic solution that satisfies the constraints but departs far from the data manifold, yielding stiff, unnatural motions.

Pseudo-observations. Replacing the adaptive pseudo-observation scheme with plain frame masking (no halo, no variance schedule) degrades FID from 0.097 to 0.880. The pseudo-observation design is therefore important for smoothly bridging temporal gaps in sparse control settings.

Sampling steps and hyperparameter robustness. We vary the number of flow steps $T \in \{20, 50, 100\}$ and find that all variants achieve exact 0.0000 constraint errors on the all-joints task; the projection mechanism is step-count agnostic. FID varies mildly from 0.103 ($T = 20$) to 0.097 ($T = 50$) to 0.095 ($T = 100$), indicating diminishing returns beyond 50 steps. Varying w_{kin} between 0.1 and 10.0 yields FID within ±0.03 of the reported value, with the largest degradation when $w_{\text{kin}} \to 0$ (recovering the Euclidean metric, consistent with the metric ablation above). These results indicate that ProjFlow is robust to its hyperparameters once they are set to a reasonable range, and that no careful per-task tuning is required to achieve exact constraint satisfaction.

7. Limitations

While ProjFlow achieves exact constraint satisfaction and strong realism, it has several limitations. First, foot-skating artifacts persist in some generated motions (Foot Skating Ratio 0.0603), as the projection step does not explicitly enforce physical contact constraints. Second, the pseudo-observation scheme relies on linear interpolation between observed keyframes; for highly non-linear motions (e.g., jumps), this can introduce erroneous guidance at the beginning of sampling. Third, ProjFlow inherits the quality of its pretrained backbone: if the base flow model generates implausible motions for a given text prompt, the projection cannot recover realism. Finally, the observation-space Cholesky factorization scales as $O(m^3)$ in the number of constrained frames m, which may be a bottleneck for very dense constraints. Moreover, the graph Laplacian metric $\mathbf{R}$ in Eq. equation 5 is fixed at inference time and does not adapt to the dynamics of the generated motion; for sequences with highly variable joint-velocity profiles, a motion-adaptive metric that scales joint stiffness by the predicted acceleration magnitude could further improve realism without sacrificing the exact constraint-satisfaction guarantee. Furthermore, ProjFlow currently assumes a known, fixed camera calibration matrix for 2D-to-3D lifting tasks; extending the framework to uncertain or multi-view settings would require marginalizing over the measurement operator, a direction we leave for future work. Our framework also processes each motion independently, which limits its applicability to multi-person interaction scenarios where constraints across bodies and inter-person contact must be handled simultaneously; adapting the constraint operator and kinematic metric $\mathbf{R}$ to such settings is an important avenue for future work.

8. Conclusion

We presented ProjFlow, a training-free projection sampler for flow-matching human motion models that enforces hard spatial constraints with zero residual error. By casting spatial motion control as a linear inverse problem and solving it in a kinematics-aware $\mathbf{R}$-norm at each sampling step, ProjFlow achieves exact constraint satisfaction while generating diverse, natural motions. Time-varying pseudo-observations with adaptive variance extend the framework to sparse inpainting settings with large temporal gaps. Evaluated on HumanML3D, ProjFlow is the only zero-shot method to achieve 0.0000 trajectory and location errors across both pelvis and all-joint control settings, while achieving competitive FID scores without any retraining. On 2D-to-3D lifting, ProjFlow achieves perfect 2D reprojection with improved 3D realism over learned baselines. We hope this work motivates further exploration of projection-based methods for structured genera-

AI-Generated

tive models beyond the motion domain. The closed-form $\mathbf{R}$-norm projection is not specific to human motion: any domain where a flow-matching model must respect hard linear constraints—image inpainting under exact pixel masks, physics-based trajectory planning with boundary conditions, or audio synthesis constrained to specified frequency bands—can leverage the same framework by substituting an appropriate domain metric into $\mathbf{R}$.

References

[1] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[2] Hyungjin Chung, Jeongsol Kim, Michael T. McCann, Marc L. Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems. In *ICLR*, 2023. 2

[3] Rishabh Dabral, Muhammad Hamza Mughal, Vladislav Golyanik, and Christian Theobalt. Mofusion: A framework for denoising-diffusion-based motion synthesis. In *CVPR*, 2023. 2

[4] Wenxun Dai et al. Motionlcm: Real-time controllable motion generation via latent consistency models. *arXiv preprint arXiv:2404.19759*, 2024. 1, 4, 6

[5] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, Dustin Podell, Tim Dockhorn, Zion English, and Robin Rombach. Scaling rectified flow transformers for high-resolution image synthesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, pages 12606–12633, Vienna, Austria, 2024. PMLR. 2

[6] Zalan Fabian, Berk Tinaz, and Mahdi Soltanolkotabi. Diracdiffusion: Denoising and incremental reconstruction with assured data-consistency. In *ICML*, 2024. PMLR 235. 2

[7] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3d human motions from text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5152–5161, 2022. 1, 4, 6

[8] Korrawe Karunratanakul, Konpat Preechakul, Supasorn Suwajanakorn, and Siyu Tang. Guided motion diffusion for controllable human motion synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 21510–21522, 2023. 1, 2, 4, 6

[9] Korrawe Karunratanakul, Konpat Preechakul, Emre Aksan, Thabo Beeler, Supasorn Suwajanakorn, and Siyu Tang. Optimizing diffusion noise can serve as universal motion priors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1334–1345, 2024. 1, 2, 4, 5, 6

[10] Jeongsol Kim, Bryan Sangwoo Kim, and Jong Chul Ye. FlowDPS: Flow-driven posterior sampling for inverse problems, 2025. 2, 3

[11] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[12] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. 2, 3

[13] Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael J. Black. AMASS: Archive of motion capture as surface shapes. In *International Conference on Computer Vision*, pages 5442–5451, 2019. 4

[14] Zichong Meng, Zeyu Han, Xiaogang Peng, Yiming Xie, and Huaizu Jiang. Absolute coordinates make motion generation easy. *arXiv preprint arXiv:2505.19377*, 2025. 1, 2, 3, 4, 6

[15] Ekkasit Pinyoanuntapong, Muhammad Usama Saleem, Korrawe Karunratanakul, Pu Wang, Hongfei Xue, Chen Chen, Chuan Guo, Junli Cao, Jian Ren, and Sergey Tulyakov. Controlmm: Controllable masked motion generation. *arXiv preprint arXiv:2410.10780*, 2024. 1, 2, 4, 6

[16] Yoni Shafir, Guy Tevet, Roy Kapon, and Amit H. Bermano. Human motion diffusion as a generative prior. In *ICLR*, 2024. 2, 4, 6

[17] Guy Tevet, Sigal Raab, Brian Gordon, Yoni Shafir, Daniel Cohen-Or, and Amit Haim Bermano. Human motion diffusion model. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2, 4, 6

[18] Yinhuai Wang, Jiwen Yu, and Jian Zhang. Zero-shot image restoration using denoising diffusion null-space model. *The Eleventh International Conference on Learning Representations*, 2023. 2

[19] Zihui Wu, Yu Sun, Yifan Chen, Bingliang Zhang, Yisong Yue, and Katherine L. Bouman. Principled probabilistic imaging using diffusion models as plug-and-play priors. In *NeurIPS*, 2024. 2

[20] Yiming Xie, Varun Jampani, Lei Zhong, Deqing Sun, and Huaizu Jiang. Omnicontrol: Control any joint at any time for human motion generation. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 2, 4, 6

[21] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion

AI-Generated

models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3813–3824, 2023. 2

[22] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu. Text-driven human motion generation with diffusion model. *IEEE TPAMI*, 2024. 2

[23] Lei Zhong, Chuan Guo, Yiming Xie, Jiawei Wang, and Changjian Li. Sketch2anim: Towards transferring sketch storyboards into 3d animation. *ACM Transactions on Graphics*, 44(4):1–15, 2025. 2, 4, 6

AI-Generated